

```
[ ] Iris_Flower_Classification.ipynb

[1] ✓ 1s
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_iris

[2] ✓ 0s
iris = load_iris()

print(iris)

... {'data': array([[5.1, 3.5, 1.4, 0.2],
                    [4.9, 3. , 1.4, 0.2],
                    [4.7, 3.2, 1.3, 0.2],
                    [4.6, 3.1, 1.5, 0.2],
                    [5. , 3.6, 1.4, 0.2],
                    ...])
```

[0.1, 4.9, 1.4, 0.2],
[4.4, 3.7, 1.3, 0.2]

```
[3] ✓ 0s df = pd.DataFrame(  
    iris.data,  
    columns=iris.feature_names  
)  
  
df['species'] = iris.target  
  
df.head()
```

...	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

Next steps: [Generate code with df](#) [New interactive sheet](#)

[4] ✓ 0s df.shape
... (150, 5)

[5] ✓ 0s df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   sepal length (cm)      150 non-null   float64
1   sepal width (cm)       150 non-null   float64
2   petal length (cm)      150 non-null   float64
3   petal width (cm)       150 non-null   float64
4   species                150 non-null   int64   
dtypes: float64(4), int64(1)
memory usage: 6.0 KB
```

```
memory usage: 6.0 KB

[6] ✓ 0s df.isnull().sum()
...
      0
sepal length (cm) 0
sepal width (cm)  0
petal length (cm) 0
petal width (cm)  0
species           0
dtype: int64

[7] ✓ 0s df.head(10)
```

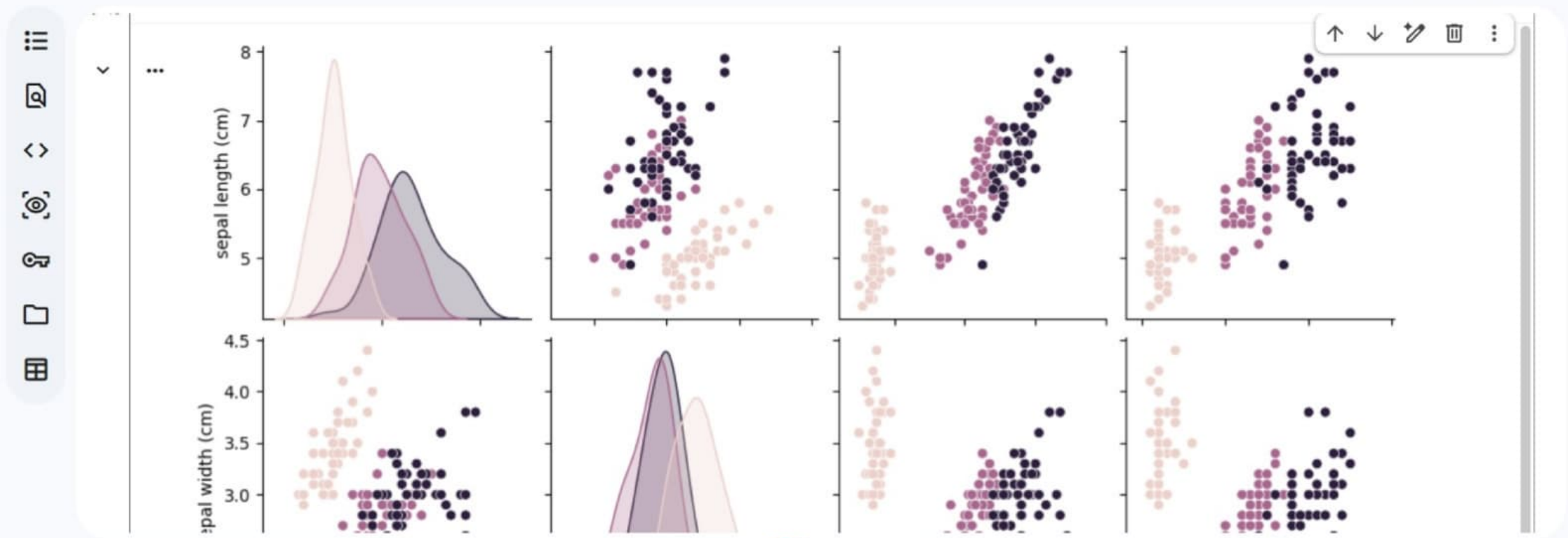
sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
-------------------	------------------	-------------------	------------------	---------

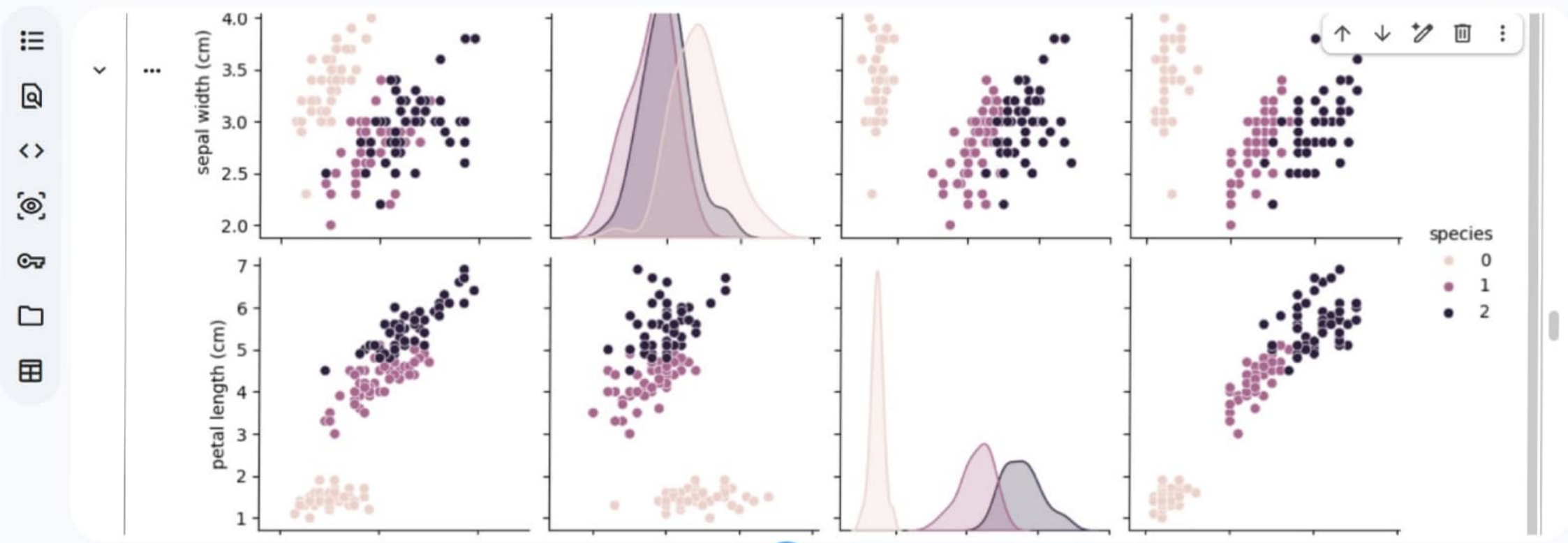
Code cell [7] showing the execution of `df.head(10)` and the resulting DataFrame output:

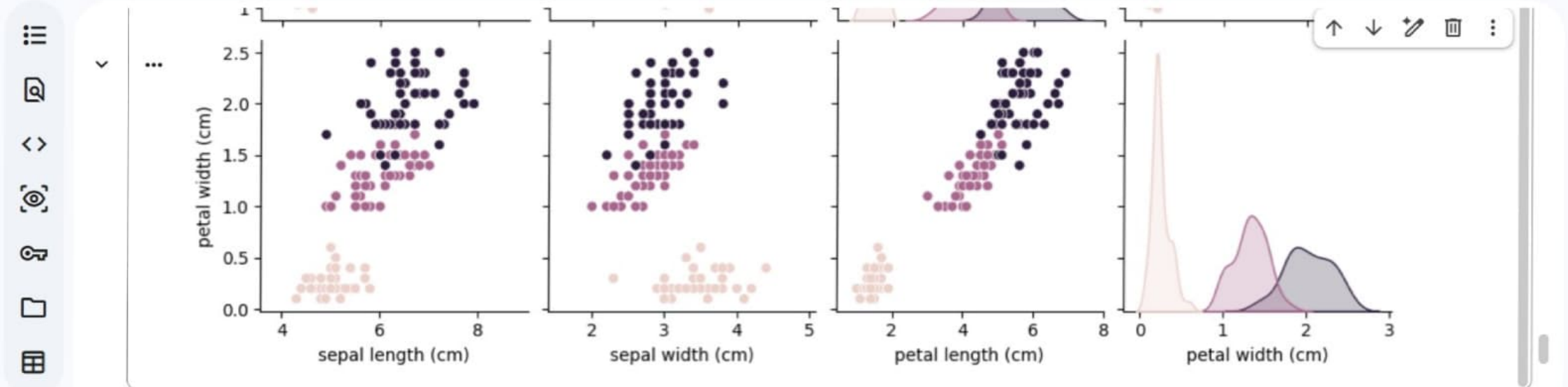
	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0
5	5.4	3.9	1.7	0.4	0
6	4.6	3.4	1.4	0.3	0
7	5.0	3.4	1.5	0.2	0
8	4.4	2.9	1.4	0.2	0
9	4.9	3.1	1.5	0.1	0

```
df.tail(10)
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
140	6.7	3.1	5.6	2.4	2
141	6.9	3.1	5.1	2.3	2
142	5.8	2.7	5.1	1.9	2
143	6.8	3.2	5.9	2.3	2
144	6.7	3.3	5.7	2.5	2
145	6.7	3.0	5.2	2.3	2
146	6.3	2.5	5.0	1.9	2
147	6.5	3.0	5.2	2.0	2
148	6.2	3.4	5.4	2.3	2
149	5.9	3.0	5.1	1.8	2







```
[11] X = df.drop('species', axis=1)
      y = df['species']
```



```
[11] ✓ 0s
X = df.drop('species', axis=1)
y = df['species']
```

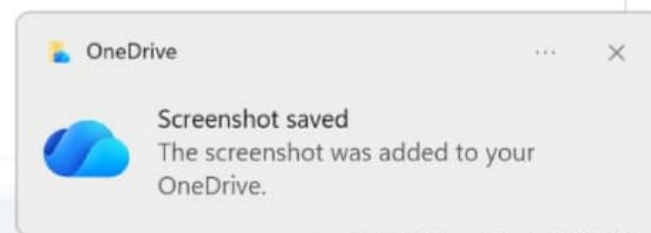
```
[12] ✓ 0s
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
[13] ✓ 0s
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier()

model.fit(X_train, y_train)
```



```
[16] ✓ 0s
classification_report(
    y_test,
    y_pred
)

...
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	9
2	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

```
[17] ✓ 0s
import pickle

pickle.dump(
    model,
    open("iris_model.pkl", "wb")
)
```

```
[14]
✓ 0s
y_pred = model.predict(X_test)

print(y_pred)

[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]

[15]
✓ 0s
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(
    y_test,
    y_pred
)

print("Accuracy:", accuracy)

Accuracy: 1.0

[16]
✓ 0s
from sklearn.metrics import classification_report

print(
    classification_report(
```

```
[17] open("iris_model.pkl", "wb")
✓ 0s )

[18] sample = [[5.1, 3.5, 1.4, 0.2]]
✓ 0s
prediction = model.predict(sample)
print(prediction)

[0]
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but RandomForestC
warnings.warn(

[19] print(
✓ 0s
    iris.target_names[prediction]
)

['setosa']
```

```
[17] open("iris_model.pkl", "wb")
✓ 0s )

[18] sample = [[5.1, 3.5, 1.4, 0.2]]
✓ 0s
prediction = model.predict(sample)
print(prediction)

[0]
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but RandomForestC
warnings.warn(

[19] print(
✓ 0s
    iris.target_names[prediction]
)

['setosa']
```